

PrintScreen Plugin v3.0.0 User Guide

Introduction

PrintScreen is an advanced screenshot and animated capture plugin for Skyrim Special Edition. It provides powerful screen capture capabilities beyond Skyrim's built-in screenshot function, supporting multiple image formats, animated captures (GIF and APNG), and sophisticated compression options.

Requirements

Before using PrintScreen, ensure you have the following installed:

- **SKSE (Skyrim Script Extender)** - Required for plugin functionality
- **SkyUI** - Required for the Mod Configuration Menu (MCM)
- **ConsoleUtil** - Required for HUD toggling during captures
- **JContainers** - Required for JSON configuration storage
- **PapyrusUtil** - Required for configuration management

Installation

1. Download the PrintScreen mod archive
2. Extract contents to your Skyrim Special Edition Data folder, or use a mod manager
3. Ensure the SKSE plugin DLL is placed in `Data/SKSE/Plugins/`
4. Launch Skyrim through SKSE

Quick Start

1. Press the configured hotkey (default: **Backspace**) to take a screenshot
2. Press the same key again during an animated capture to cancel it
3. Access the MCM menu under "PrintScreen" to configure settings

Image Formats

PrintScreen supports the following image formats:

Format	Extension	Best For
PNG	.png	High-quality screenshots with lossless compression
APNG	.png	Animated captures with full color depth
BMP	.bmp	Uncompressed bitmap images
TIF	.tif	Professional archival with various compression options
JPG	.jpg	Smaller file sizes with adjustable quality
GIF	.gif	Simple static screenshots (256 colors)
AGIF	.gif	Animated captures with frame optimization
DDS	.dds	DirectX texture format for game asset creation

Configuration Parameters

General Settings

Path

- Description: The folder where screenshots will be saved
- Default: `C:/Pictures`
- Format: Full Windows path (e.g., `C:/Users/YourName/Pictures/Skyrim`)
- Note: The plugin will validate the path and attempt to create the directory if it doesn't exist

Image Type

- Description: The format for captured images
- Options: PNG, APNG, BMP, TIF, JPG, GIF, AGIF, DDS
- Default: PNG

Take Photo Key

- Description: The keyboard key to trigger a capture
- Default: Backspace (key code 14)
- Note: Press during an active animated capture to cancel it

Automatic Menu Removal

- Description: Automatically hide HUD elements during capture
- Values: On/Off
- Default: On

Save/Restore Configuration

- Description: Persist settings to JSON file between game sessions
 - Values: On/Off
 - Default: On
-

JPG Parameters

JPG Compression (Quality)

- Description: Image quality level for JPEG format
 - Range: 0-100
 - Default: 85
 - Note: Higher values mean better quality but larger files
-

TIF Parameters

TIF Mode (Compression)

- Description: Compression algorithm for TIFF files
 - Options:
 - **UNCOMPRESSED** - No compression, largest files
 - **RLE** - Run-length encoding, good for simple images
 - **LZW** - Lempel-Ziv-Welch, general-purpose lossless
 - **ZIP** - Deflate compression, good compression ratio
 - Default: UNCOMPRESSED
-

DDS Parameters

DDS Mode (Compression)

- Description: Block compression format for DDS textures
 - Options:
 - **UNCOMPRESSED** - No compression
 - **BC1** - RGB, 4:1 compression (no alpha)
 - **BC2** - RGBA with explicit alpha
 - **BC3** - RGBA with interpolated alpha
 - **BC4** - Single-channel grayscale
 - **BC5** - Two-channel (normal maps)
 - **BC6H** - HDR RGB images
 - **BC7_SLOW** - Highest quality, slowest encoding
 - **BC7_NORMAL** - Balanced quality/speed
 - **BC7_FAST** - Faster encoding, slightly lower quality
 - Default: BC1
-

Animated Capture Parameters (AGIF/APNG)

These parameters apply to both Animated GIF (AGIF) and Animated PNG (APNG) formats.

Duration

- Description: Total capture time in seconds
- Range: 0-30 seconds
- Default: 30 seconds
- Note: Press the capture key again to stop early

FPS (Frames Per Second)

- Description: Frame rate for animated captures
- Range: 1-30 FPS
- Default: 10 FPS
- Note: Higher values create smoother animations but larger files

Loop Count

- Description: Number of times the animation repeats
- Range: 0-10

- Default: 0 (infinite loop)
- Values:
 - 0 = Loops forever
 - 1+ = Plays the specified number of times then stops

Compression

- Description: Enable differential frame encoding
- Values: 0 (disabled) or 1 (enabled)
- Default: 1
- Note: When enabled, only changed portions of each frame are stored

Optimize (Delta Encoding)

- Description: Advanced optimization mode for animated captures
 - Values: 0 or 1
 - Default: 1
 - Details:
 - **0 = Region Extraction Only:** Only the bounding box of changed pixels is stored
 - **1 = True Delta Encoding:** Unchanged pixels become transparent, enabling superior compression
 - Note: True delta encoding typically produces files 30-60% smaller than region extraction alone
-

MCM Configuration Guide

Access the PrintScreen configuration through:

1. Press Escape to open the game menu
2. Navigate to "Mod Configuration"
3. Select "PrintScreen"

Settings Page Layout

The MCM provides the following controls organized by relevance to your selected image type:

1. **Header:** Displays current version number
2. **Path Input:** Enter your save location
3. **Image Type Menu:** Select capture format

4. **Automatic Menu Removal Toggle:** HUD hiding preference
5. **Take Photo Key:** Assign your capture hotkey
6. **Save/Restore Configuration:** JSON persistence toggle
7. **Format-Specific Options:** Sliders and menus that appear based on your selected format

Dynamic Option Visibility

The MCM dynamically shows/hides options based on your selected image type:

- **JPG selected:** JPG Compression slider appears
 - **TIF selected:** TIF Mode menu appears
 - **DDS selected:** DDS Mode menu appears
 - **AGIF/APNG selected:** All animated parameters appear (Duration, FPS, Loop Count, Compression, Optimize)
 - **PNG/BMP/GIF selected:** Only basic options shown
-

INI Configuration

Advanced users can configure logging and performance settings via the INI file.

Location: `Documents/My Games/Skyrim Special Edition/SKSE/PrintScreen.ini`

Logging Section

```
ini

[Logging]
LogLevel=INFO      ; NONE, ERROR, WARN, INFO, DEBUG, TRACE
ConsoleOutput=true  ; Output to Visual Studio debugger
FileOutput=true     ; Write to log file
ShowTimestamps=true ; Include timestamps in log entries
MaxLogFileSizeMB=8  ; Maximum log file size before rotation
```

Performance Section

```
ini

[Performance]
ParallelCompression=true ; Use multiple threads for compression
CompressionThreads=0     ; Number of threads (0 = auto-detect)
```

Capture Section

ini

[Capture]

LogCaptureProgress=false ; Log frame-by-frame progress

LogTimingInfo=false ; Log timing information

Usage Tips

Taking Single Screenshots

1. Configure your desired format and settings in MCM
2. Position your character and camera
3. Press your capture hotkey
4. Wait for the "Screenshot succeeded" notification

Creating Animated Captures

1. Select AGIF or APNG as your image type
2. Set your desired Duration, FPS, and optimization settings
3. Press your capture hotkey to begin recording
4. The HUD will automatically hide (if enabled)
5. Wait for the duration to complete, or press the hotkey again to stop early
6. Wait for the encoding process to complete

Optimizing File Sizes

For the smallest animated files:

- Enable Compression (set to 1)
- Enable Optimize/Delta Encoding (set to 1)
- Use lower FPS (10 is usually sufficient)
- Capture scenes with minimal movement when possible

For best quality:

- Use APNG format (full 32-bit color)
- Higher FPS for smooth motion

- PNG format for static screenshots
-

Troubleshooting

"Screenshot failed" error

- Verify your save path exists and is writable
- Check that you have sufficient disk space
- Review the log file at `Data/SKSE/Plugins/Printscreen.log`

Animated capture won't start

- Ensure a previous capture has completed
- Use the ForceReset function via console if stuck
- Try reloading your save

Poor animation quality

- Increase FPS setting
- Use APNG instead of AGIF for full color depth
- Reduce Duration if files are too large

Large file sizes

- Enable both Compression and Optimize options
 - Lower the FPS setting
 - Use JPG format for static screenshots where quality loss is acceptable
-

Log File Location

Debug and error information is written to:

`Data/SKSE/Plugins/Printscreen.log`

Review this file when troubleshooting issues.

Version History

v3.0.0

- Added true delta encoding for significantly smaller animated files
 - Improved state management between Papyrus and C++ layers
 - Added race condition protection for capture operations
 - Enhanced cancellation support for animated captures
 - Added APNG format support with differential encoding
 - Comprehensive logging system with configurable levels
-

Credits

Author: William G Lea

Technologies Used:

- SKSE (Skyrim Script Extender)
- CommonLibSSE-NG
- DirectXTex
- Windows Imaging Component (WIC)
- Windows Desktop Duplication API